Mark schemes

Q1.

(a) $\frac{dx}{dt} = 4 \quad \frac{dy}{dt} = -\frac{1}{2t^2}$

differentiate. 4; at^{-2} seen

M1

both derivatives correct

A1

$\frac{dy}{dx} = -\frac{1}{2t^2} \times \frac{1}{4}$

use chain rule candidates' $\frac{dy}{dt} / \frac{dx}{dt}$

M1

$t = \frac{1}{2} \quad \frac{dy}{dx} = -\frac{1}{2}$

CSO

A1

ALTERNATIVE METHOD:

$y = \frac{2}{x-3} - 1$ and differentiate

differentiate expression in y and x

M1

$\frac{dy}{dx} = \frac{-2}{(x-3)^2}$

correct

A1

$x = 5$

$\frac{dy}{dx} = \frac{-2}{(5-3)^2}$

find and therefore use x (and y)

m1

$\frac{dy}{dx} = -\frac{1}{2}$

A1

4

(b) gradient of normal = 2

F if gradient ≠ ±1

B1F

$(x, y) = (5, 0)$ $\quad \frac{y}{x-5} = 2$

calculate and use (x, y) on normal

M1

F on gradient of normal ACF

A1F

3

(c) $x - 3 = 4t$ or $y + 1 = \frac{1}{2t}$

or $t = \frac{x-3}{4}$ or $\frac{1}{t} = 2(y+1)$

B1

$(x - 3)(y + 1) = 2$

eliminate t; allow one error

M1

$y = \frac{1}{\frac{2(x-3)}{4}} - 1$ ACF

accept

SC allow marks for part (c) if done in part (a)

A1

3

[10]